Vision Error Observatory (VEO) Diagnostic Pipeline

Six-Stage Framework for Metric Error Localization, Evidence Extraction, and Failure Attribution

STAGE 1: INPUT & ALIGNMENT

- RGB Image: $I \in \mathbb{R}^{H \times W \times 3}$
- Predicted Depth: $P \in \mathbb{R}^{H \times W}$
- Ground Truth: $D_{\text{gt}} \in \mathbb{R}^{H \times W}$

Inverse-Depth Affine Fit:

$$\min_{s,b} \sum (sP + b - 1/D_{\text{gt}})^2$$

Strict Polarity: $s > 0$ **(244/244)**

Masking: Reject zeros & NaN

STAGE 2: REGION EXTRACTION

Relative Error Surface:

$$E_{\text{rel}}(x,y) = \frac{|D_{\text{pred}} - D_{\text{gt}}|}{D_{\text{gt}}}$$

Adaptive 85th %ile Threshold:

$$\tau_{\text{error}} = \text{Percentile}_{85}(E_{\text{rel}})$$

Connected Components (8-conn)

Filter Area: $|R_k| \geq 25\,\text{px}$

Yields 5,236 NYU regions

STAGE 3: EVIDENCE SIGNALS

1. Texture Var	Var(I_gray)
2. Edge Density	Canny(100,200)/255
3. Brightness	Mean regional gray
4. Contrast	Std regional gray
5. Specular Glare	HSV V>200, S<40
6. Aspect Ratio	max(W,H)/min(W,H)
7. Depth Step	||grad D_gt|| on bnd

STAGE 4: CONTINUOUS SCORING & CALIBRATED AMBIGUITY GATE

Continuous Linear Ramps [0, 1]:

- score_textureless = clip01(1 - texture_var / !
- score_boundary = clip01(edge_density * 3.0
- score_lighting = clip01(1 - contrast / 60.0) if
- score_thin = clip01((aspect_ratio - 3.0) / 5.0
- score_reflection = clip01(highlight_ratio * 5.
- score_occlusion = clip01(depth_discontinuit

Ambiguity Decision Gate

Leading Scores: $s_{(1)} \geq s_{(2)}$

Condition: $s_{(1)} < 0.40$ **OR**

$\text{margin} = s_{(1)} - s_{(2)} < 0.10$

True -> 'ambiguous' (Low Conf)

False -> Top Cause (High/Med Conf)

Prevents False Certainty

STAGE 5: DOWNSTREAM AUDIT

Verified Actionable Utility

1. Architectural Fingerprints:
   ViT: textureless & boundary
   Conv: reflection & lighting
2. Counterfactual Testing:
   Noise injection flips label
3. Selective Boundary Filter:
   +0.354% Regional AbsRel
   +0.162% Regional RMSE
   across 207 test scenes